Plan of the Day

- Budgeting Game Check In
- Portfolio Activity (Tuesday: Do, Thursday: Discuss)
- Investing Basics
- Investment growth is not linear…think probability!
- Mutual Fund Activity (time permitting)

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CWRU
MGMT 205
Essentials of Personal Finance

Basics of Investing

Presented by

Mary Sasmaz, PhD, CPA, CFP
Department of Accountancy
Weatherhead School of Management
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Agenda

- Quick Recap on Time Value of Money
- Investing for Retirement
  - Strategy
  - Investment Basics
  - Diversification

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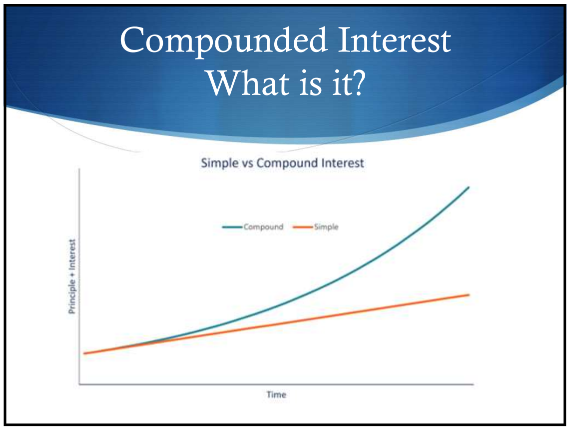
Compounded Interest
What is it?
Simple vs Compound Interest
Compound
Simple
Principle + Interest
Time

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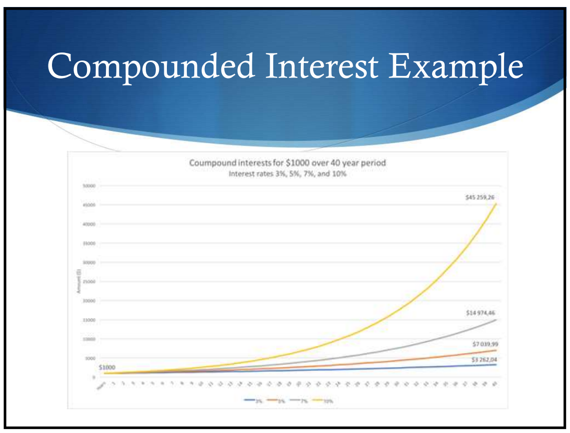
Compounded Interest Example
Coumpound interests for $1000 over 40 year period
Interest rates 3%, 5%, 7%, and 10%
$45 259,26
$14 974,46
$7 039,99
$3 262,04
$1000

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Investing for Retirement

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Investments

- Investment strategy will differ from emergency fund or planned expense funds
  - Long-term
  - Regular Basis
  - Deal with Risk
- Risk vs Return - Note the retirement savings plan discussed previously assumes an average return/growth rate of 7.5% - requires taking on risk. (You need 4.5% after inflation).
- Many options – You need a mix
  - Stocks (i.e. equities)
  - Bonds
  - Funds
    - Mutual Funds / ETFs
    - Bond Funds
    - Index Funds

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Investments
Stocks (Equities)

- A share in the ownership of a company. If it's traded on the stock market, then the company is a "public" or "publicly owned"
- You own a little tiny piece of the company. You benefit from the performance of the company – stock value change and dividends, if any.

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Investments
Bonds

- A "fixed income" instrument that is basically a loan from the investor (lender) to the borrower (corporation / government).
- Ex. Company borrows money from you and promises to pay you interest while borrowing it and payback the principal amount at maturity.
- Bonds are priced based on current market rates, coupon (interest rate), credit rate of company, and borrowing period of time.

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Diversification

- Investment Growth is not exactly linear!

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S&P 500 – historical chart

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Asset Class Returns

2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	Total
EM 79.0%	REIT 28.0%	REIT 8.3%	REIT 19.7%	Sm Cap 38.8%	REIT 28.0%	REIT 2.8%	Sm Cap 21.3%	EM 37.8%	Cash [illegible]	Lg Cap 31.5%	Sm Cap 20.0%	REIT 41.3%	Cash [illegible]	Lg Cap 26.3%	Lg Cap 610.9%
HY Bnd 57.5%	Sm Cap 26.9%	HG Bnd 7.8%	EM 18.6%	Lg Cap 32.4%	Lg Cap 13.7%	Lg Cap 1.4%	HY Bnd 17.5%	Int'l 25.6%	HG Bnd 0.0%	REIT 28.7%	EM 18.7%	Lg Cap 28.7%	HY Bnd -11.2%	Int'l 18.9%	Sm Cap 398.3%
Int'l Stk 32.5%	EM 19.2%	HY Bnd 4.4%	Int'l Stk 17.9%	Int'l Stk 23.3%	AA 6.9%	HG Bnd 0.6%	Lg Cap 12.0%	Lg Cap 21.8%	HY Bnd -2.3%	Sm Cap 25.5%	Lg Cap 18.4%	Sm Cap 14.8%	HG Bnd -13.0%	Sm Cap 16.9%	REIT 369.4%
REIT 28.0%	HY Bnd 15.2%	Lg Cap 2.1%	Sm Cap 16.4%	AA 11.5%	HG Bnd 6.0%	Cash [illegible]	EM 11.6%	Sm Cap 14.7%	REIT -4.0%	Int'l Stk 22.7%	AA 9.8%	Int'l Stk 11.8%	Int'l Stk -14.0%	HY Bnd 13.5%	HY Bnd 270.3%
Sm Cap 27.2%	Lg Cap 15.1%	AA 0.3%	Lg Cap 16.0%	HY Bnd 7.4%	Sm Cap 4.9%	Int'l Stk -0.4%	REIT 8.6%	AA 14.6%	Lg Cap -4.4%	AA 18.9%	Int'l Stk 8.3%	AA 10.9%	AA -16.5%	AA 12.8%	AA 196.4%
Lg Cap 26.5%	AA 13.5%	Cash [illegible]	HY Bnd 15.6%	REIT 2.9%	HY Bnd 2.5%	AA -1.3%	AA 7.2%	REIT 8.7%	AA -5.6%	EM 18.9%	HY Bnd 7.5%	HY Bnd 5.4%	Lg Cap -18.1%	REIT 11.4%	Int'l 193.3%
AA 24.6%	Int'l Stk 8.2%	Sm Cap -4.2%	AA 12.2%	Cash [illegible]	Cash [illegible]	Sm Cap -4.4%	HG Bnd 2.7%	HY Bnd 7.5%	Sm Cap -11.0%	HY Bnd 14.4%	HG Bnd 6.1%	Cash [illegible]	EM -19.7%	EM 10.3%	EM 173.2%
HG Bnd 5.9%	HG Bnd 6.5%	Int'l Stk -11.7%	HG Bnd 4.2%	HG Bnd -2.0%	EM -1.8%	HY Bnd -4.6%	Int'l Stk 1.5%	HG Bnd 3.5%	Int'l Stk -13.4%	HG Bnd 8.7%	Cash [illegible]	HG Bnd -1.5%	Sm Cap -20.4%	HG Bnd 5.5%	HG Bnd 46.8%
Cash [illegible]	Cash [illegible]	EM -18.2%	Cash [illegible]	EM -2.3%	Int'l Stk -4.5%	EM -14.6%	Cash [illegible]	Cash [illegible]	EM -14.3%	Cash [illegible]	REIT -5.1%	EM -2.2%	REIT -25.0%	Cash [illegible]	Cash [illegible]

Abbr.	Asset Class – Index	Annual	Best	Worst
Lg Cap	Large Cap Stocks – S&P 500 Index	13.97%	32.4%	-18.1%
Sm Cap	Small Cap Stocks – Russell 2000 Index	11.30%	38.8%	-20.4%
Int'l Stk	International Developed Stocks – MSCI EAFE Index	7.44%	32.5%	-14.0%
EM	EM Stocks – MSCI Emerging Markets Index	6.93%	79.0%	-19.7%
REIT	REITs – FTSE NAREIT All Equity Index	10.86%	41.3%	-25.0%
HG Bnd	High Grade Bonds – Bloomberg Barclays U.S. Agg Index	2.59%	8.7%	-13.0%
HY Bnd	High Yield Bonds – ICE BofA US High Yield Index	9.12%	57.5%	-11.2%
Cash	Cash – S&P U.S. Treasury Bill 0-3 Mth Index	0.85%	5.1%	0.0%
AA	Asset Allocation Portfolio*	7.51%	24.6%	-16.5%

Past performance does not guarantee future returns. The historical performance shows changes in market trends across several asset classes over the past fifteen years. Returns represent total annual returns (reinvestment of all distributions) and does not include fees and expenses. The investments you choose should reflect your financial goals and risk tolerance. For assistance, talk to a financial professional. All data are as of 12/31/23.

*Asset Allocation Portfolio is 15% large cap stocks, 15% international stocks, 10% small cap stocks, 10% emerging market stocks, 10% REITs, 40% high-grade bonds, and annual rebalancing.

Source: novelinvestor.com

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Diversification

- Don't put all your eggs in one basket
  - Diversify your portfolio to reduce your investment risk
    - Age and Stage of Life will be factors in level of diversification
  - Basic understanding
    - Equities – Risky
    - Bonds – Less risky than equities, (at least in short-term)
    - Cash – No risk

- Equities: Different industries, different size companies, different countries
- Bonds: Different industries, credit risks, maturities, countries
- The best plan is the one that lets you sleep at night. You must keep risk level within your own "comfort zone."

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Diversification

- Advice will vary between advisors
- Advisor 1: Rule of Thumb – The Age Formula:
  - Age this year = 43
    - 110 – 43 = 67% in Equities (33% in Bonds and Cash)
  - Age this year = 58
    - 110 – 58 = 52% in Equities (48% in Bonds and Cash)
- Advisor 2: Rule of Thumb (Stock / Bond & Cash)
  - 50/50 until age 60, then switch to 40/60
- Some computerized systems build an asset allocation to you based on responses to a risk level questionnaire (ex. Betterment – Robo advisor)risk-level

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Investments
Mutual Funds / Bond Funds

- A "basket of investment securities" that is put together by an Investment Company
- Mutual/Bond Fund sells/buys shares to/from investors who want to invest in the fund (i.e. basket).
- Value changes based on the value of the securities in the fund/basket
- Can be widely diversified or concentrated in number of stocks/bonds and industry/ies.
- Minimum investment amount differs for each fund

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Short-term vs Long-term Results

One (1) Year Annual Returns

Portfolio Mix	Best Annual Return	Occurred In Year	Worst Annual Return	Occurred In Year	1926 to 2013 Annual Average Return	Over 88 Yrs. Times Positive
100 % S&P 500 Index	53.99%	1933	-43.34%	1931	12.05%	64
50% S&P 500 - 50% Govt. Bonds	34.71%	1995	-24.70%	1931	8.88%	69
100% Long-term Govt. Bonds	40.36%	1982	-14.90%	2009	5.91%	65

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Short-term vs Long-term Results

Twenty (20) Year Rolling Period Returns

Portfolio Mix	Best 20 Year Return	20 Year Period	Worst 20 Year Return	Occurred In Year	20 Year Annual Average Return	Over 69 Periods Times Positive
100 % S&P 500 Index	17.88%	1980-99	3.11%	1929-1948	11.18%	69
50% S&P 500 - 50% Govt. Bonds	14.75%	1979-1998	4.60%	1929-1948	8.71%	69
100% Long-term Govt. Bonds	12.09%	1982-2001	0.01%	1950-1969	5.53%	69

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However, the probability of a positive return in the S&P 500 improves as holding period lengthens:

HOLDING PERIOD	ODDS OF A POSITIVE RETURN IN S&P 500
1 month	59%
1 year	69%
5 years	79%
10 years	88%
20 years	100%

DATA SOURCE: BLOOMBERG, CRESTMONT RESEARCH, YARDENI RESEARCH. SHOWN IS THE ODDS OF A POSITIVE RETURN IN THE S&P 500 OVER DIFFERENT TIME PERIODS BASED ON DATA COLLECTED BETWEEN 1928 AND 2023.

The S&P 500 has been a profitable investment over every rolling 20-year period since 1928. That means owning an S&P 500 index fund for at least two decades has always been a profitable investment strategy.

Source: Motley Fool

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Things to Remember
Long-term Perspective

- These funds are invested for retirement, not accessed before then.
- Don't look at your account balance every day – you'll go crazy!
  - **Trend in value** is more important in today's value. (10 to 30 years)
- Long-term growth – need equities/stocks
- Bonds and cash –lower the risk of over-all portfolio
- Don't freak when market drops, market lows are a good time to buy, but engaging in "market timing" is very difficult over the long period.
- Do your homework – don't invest in something just because someone else told you to – do your own research first.

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Yahoo Finance Mutual Fund Research Activity

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Next Time…..

- Active versus Passive Management
- Building a Portfolio
- Target Date Retirement Funds

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